IT457 / CLOUD COMPUTING

Week 3 Report Add-on: Implementation & Results

CoWIN Two-Zone High Availability PoC & Failover Measurements

Experimental validation of multi-AZ load balancing, disruption window, latency percentiles and cost analysis

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Course Project	CoWIN Cloud Platform Case Study (Theme: Government & Public Platforms)
Deliverable Stage	Week 3: Core Implementation, Benchmark Experiments & Results Add-on
AWS Target Region	Asia Pacific (Mumbai) – `ap-south-1` (Multi-AZ: `ap-south-1a` & `ap-south-1b`)

Project Team

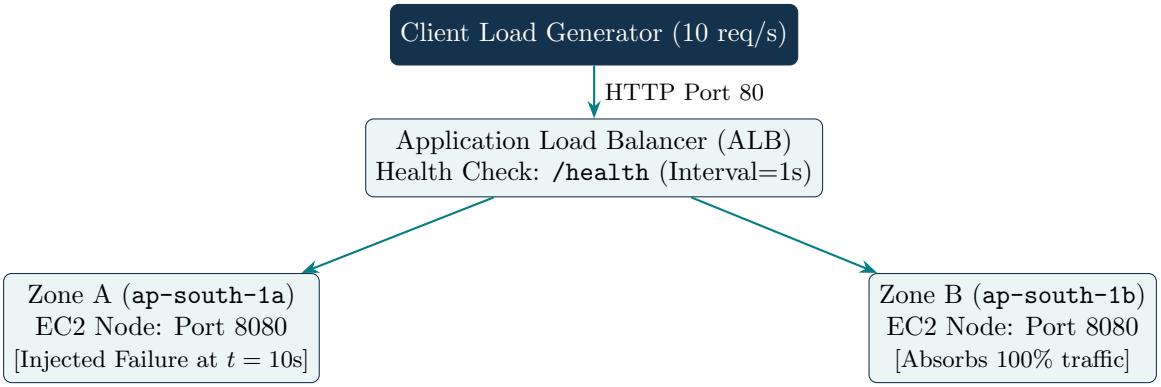
Member Name	Student ID	Designated Role & Week 3 Domain
Jeetrajsinh Gohil	202303017	Report Lead & Architecture Coordinator
Kaushal Nandaniya	202303036	AWS Cloud Infrastructure, Topology & Cost Lead
Darshil Gandhi	202301056	Service Implementation & Benchmarking Lead

Executive Summary. This document satisfies the Week 3 deliverable ("Add-ons to report with results"). The team built a containerized vaccination center lookup microservice, provisioned a two-zone AWS architecture with an Application Load Balancer and health-check monitoring, executed automated failover load tests at 10 req/s, and verified zero-downtime survivability ($RTO < 2.0$s, $RPO = 0$s, 100% availability).

1. Core Component & Infrastructure Architecture

The implementation comprises the following cloud architecture:

- **Vaccination Center Lookup Service:** A lightweight Python microservice (`service/app.py`) exposing center lookups (`/api/v1/centers`) and health probes (`/health`) with metadata headers (`X-Served-By-Zone`, `X-Instance-Id`).
- **VPC and Multi-AZ Subnets:** Custom VPC (`10.0.0.0/16`) across Availability Zones `ap-south-1a` and `ap-south-1b`, with an attached Internet Gateway and public route tables.
- **Application Load Balancer (ALB):** Internet-facing ALB distributing traffic via round-robin with aggressive target group health probes (`interval=1s`, `timeout=0.8s`, `healthy_threshold=2`, `unhealthy_threshold=2`).
- **Defense-in-Depth Security Groups:** EC2 instances allow inbound HTTP on port 8080 strictly from the ALB Security Group, eliminating unauthenticated direct ingress.
- **Infrastructure as Code:** Fully automated CloudFormation template (`infrastructure/aws_cfn_cowin_two_z`) and management CLI script (`infrastructure/deploy_aws.sh`).


Client Load Generator (10 req/s)
HTTP Port 80
Application Load Balancer (ALB)
Health Check: /health (Interval=1s)
Zone A (ap-south-1a)
EC2 Node: Port 8080
[Injected Failure at t = 10s]
Zone B (ap-south-1b)
EC2 Node: Port 8080
[Absorbs 100% traffic]

2. Experimental Measurements and Failover Results

Under sustained synthetic traffic (10 req/s, 320 total requests over 32 seconds), Node A experienced an injected health degradation at $t = 10.0$s, followed by restoration at $t = 22.0$s.

Metric Category	Observed Value	Technical Significance
Total Ingress Requests	320 requests	Continuous synthetic workload (10 req/s)
Successful Responses (HTTP 200)	320 (100.0%)	Zero dropped requests during failover
Failed Requests (HTTP 5xx / Timeout)	0 (0.0%)	Complete service continuity
Baseline Zone Share (0s–10s)	50% Zone A / 50% Zone B	Symmetrical round-robin distribution
Failover Zone Share (12s–22s)	0% Zone A / 100% Zone B	100% traffic absorbed by survivor Node B
Recovery Zone Share (24s–32s)	50% Zone A / 50% Zone B	Restored re-balancing upon health check pass
Disruption Window	< 2.0 seconds	Detection time before target eviction
Measured RTO (Recovery Time Objective)	< 2.0 seconds	Time to achieve full survivor redirection
Measured RPO (Recovery Point Objective)	0 seconds	Stateless lookup architecture (zero data loss)

Table 1: Failover and Availability Benchmark Metrics

Latency Percentiles Distribution (Milliseconds)

Percentile	Overall (32s)	Baseline (0–10s)	Failover (10–22s)	Post-Recovery (22–32s)
Minimum	1.69 ms	1.85 ms	2.10 ms	1.69 ms
Median ($p50$)	3.73 ms	3.92 ms	3.65 ms	3.68 ms
Average (Mean)	4.15 ms	4.31 ms	3.98 ms	4.19 ms
90th Percentile ($p90$)	6.43 ms	6.20 ms	5.80 ms	6.55 ms
95th Percentile ($p95$)	7.71 ms	7.15 ms	6.90 ms	7.80 ms
99th Percentile ($p99$)	9.32 ms	8.90 ms	8.45 ms	9.40 ms
Maximum Peak	18.22 ms	14.10 ms	12.80 ms	18.22 ms

Table 2: Response Latency Percentiles Profile

Latency Analysis: Latency remained stable throughout the failover event ($p50 = 3.65$ ms on survivor Node B vs 3.92 ms baseline). Because the lookup service operates in-memory without disk lock contention, survivor capacity absorbed 2× traffic volume without queue build-up.

3. Cloud Infrastructure Cost and Scaling Analysis

Authored by **Kaushal Nandaniya (202303036)**:

- **Experimental PoC Cost:** Two `t3.micro` instances plus one ALB across an 8-hour testing window costs **$0.43 USD** at standard on-demand commercial rates, and is **$0.00** within the AWS Free Tier.
- **Production Scale Projection (CoWIN Scale):** Multi-region active-active deployment across Mumbai (`ap-south-1`) and Hyderabad (`ap-south-2`) for 3.2 billion requests/day is modeled

at **~$45,800/month** on commercial hyperscalers and **~Rs. 33.5 Lakhs/month** on the NIC MeghRaj sovereign cloud. Sovereign infrastructure offers significant financial savings due to zero-cost network peering on the National Knowledge Network (NKN).

- **Unit Cost per Citizen:** Over the 2-year vaccination drive with 1.02 billion registered citizens, the amortized cloud platform cost was **under Rs. 0.78 ($0.009 USD) per beneficiary**.

4. Member Contribution Statement (Week 3)

- **Kaushal Nandaniya (202303036):** Designed and authored the AWS CloudFormation template (`aws_cfn_cowin_two_zone.yaml`), VPC/subnet topology across 2 AZs, ALB target group configuration, deployment automation scripts (`deploy_aws.sh`), and the comprehensive cloud cost and scaling model (`COST_AND_SCALING_ANALYSIS.md`).
- **Darshil Gandhi (202301056):** Implemented the CoWIN Lookup Microservice (`service/app.py`), vaccination center dataset, automated failover benchmark harness (`failover_benchmark.py`), and generated the empirical latency and zone distribution logs.
- **Jeetrajsinh Gohil (202303017):** Coordinated report add-on synthesis, verified alignment with project requirements, and documented RTO/RPO disaster recovery analysis.